

Technology #02: Adaptive Valve Profile for Postural ICP Regulation

Technology Domain: Fluid Mechanics + Adaptive Valve Design

Mechanism: CSF shunt valve whose opening pressure profile adapts to ICP trend rather than fixed cracking pressure; reduces postural pressure excursions

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: Adaptive response time exceeds postural transient duration (cannot respond within seconds)

Next Experiment: Variable-area valve prototype (passive, no actuator)

Buyer Receives: Adaptive valve concept + mechanism description; Controller algorithm specification (MODELLED, not production-firmware); Variable-area orifice governing equation + sensitivity analysis; V0 bench prototype BOM (COTS sensor + custom valve body + candidate actuators); This engineering dossier with external precedent + failure analysis + build plan

Buyer Must Build: Production valve design (dimensioned drawings, tolerances, material selection frozen); Actuator technology selection and validation; Production controller firmware + software V&V; per IEC 62304; Manufacturing process for variable-area valve; Regulatory submission (likely 510(k) with substantial equivalence, or De Novo if no predicate); Clinical validation evidence (IDE trial); Long-term reliability testing (ISO 14708-1 for active implants)

Transfer Posture: Sponsored Validation

Target Buyer: Neuroshunt manufacturer with active valve program

CONFIDENTIAL. See full engineering dossier for details.